

Session4_hw

AUTHOR

Máté Antal

```
library(readr)
library(readxl)
library(fpp3)
```

— Attaching packages

fpp3 1.0.3 —

✓ tibble	3.3.1	✓ tsibble	1.2.0
✓ dplyr	1.2.1	✓ tsibbledata	0.4.1
✓ tidyr	1.3.2	✓ ggtime	1.0.0
✓ lubridate	1.9.5	✓ feasts	0.5.0
✓ ggplot2	4.0.3	✓ fable	0.5.0

— Conflicts —

fpp3_conflicts —

* lubridate::date()	masks base::date()
* dplyr::filter()	masks stats::filter()
* tsibble::intersect()	masks base::intersect()
* tsibble::interval()	masks lubridate::interval()
* dplyr::lag()	masks stats::lag()
* tsibble::setdiff()	masks base::setdiff()
* tsibble::union()	masks base::union()

04_C_TSGraphs_Lagplots_Autocorrelation

exercise 1 - matching 1- b - the autocorrelations are positive but gradually fade, showing persistence without strong seasonality, which we can see on the graph

2 - a - the ACF has a clear spike around lag 12, meaning yearly seasonality in monthly data

3 - d - the ACF stays very high and decreases slowly, which is what usually happens for strong upward trends, and the ACF crosses the blue line by far which show very strong significance (very strong trend in graph)

4 - c - the ACF alternates between positive and negative values, matching the cyclic rises and falls in the series.

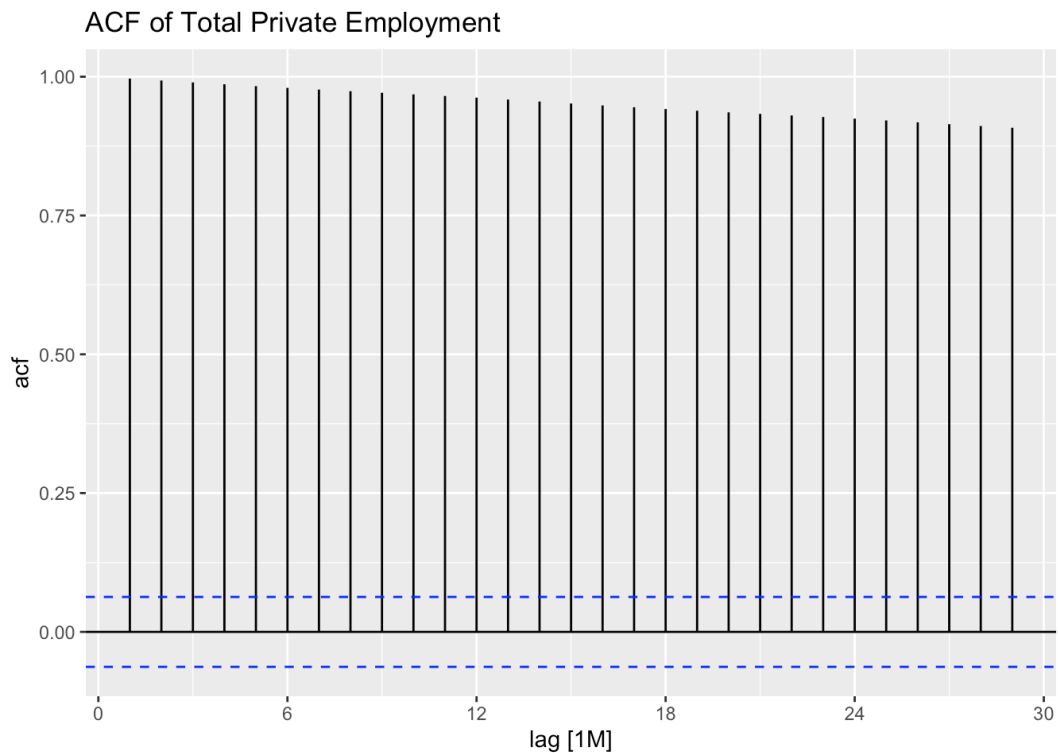
```
total_private <- us_employment |>
```

```
filter(Title == "Total Private")

total_private |>
  autoplot(Employed) +
  labs(
    title = "Total Private Employment",
    y = "Employment"
  )
```



```
total_private |>
  ACF(Employed) |>
  autoplot() +
  labs(
    title = "ACF of Total Private Employment"
  )
```



Total private employment shows a very strong long-term upward trend with some downturns during recessions like 2008, and the ACF decays slowly because nearby employment values are persistent over time, so even a few months later the data will be quite similar, but the earliest data being the best predictor.

Exercise 3

```
PBS |>
  filter(ATC2 == "H02") |>
  select(ATC2, Month, Cost, everything()) |>
  arrange(Month)
```

```
# A tibble: 816 x 9 [1M]
# Key:      Concession, Type, ATC1, ATC2 [4]
   ATC2      Month  Cost Concession  Type      ATC1  ATC1_desc
ATC2_desc Scripts
   <chr>    <mtm>  <dbl> <chr>      <chr>    <chr> <chr>
<chr>    <dbl>
1 H02    1991 Jul 317384 Concessional Co-paym... H      Systemic...
CORTICOS... 63261
2 H02    1991 Jul  82929 Concessional Safety ... H      Systemic...
CORTICOS... 12211
3 H02    1991 Jul  20178 General      Co-paym... H      Systemic...
CORTICOS... 4013
4 H02    1991 Jul   9304 General      Safety ... H      Systemic...
CORTICOS... 2101
```

```

5 H02    1991 Aug 269891 Concessional Co-paym... H    Systemic...
CORTICOS... 53528
6 H02    1991 Aug 100602 Concessional Safety ... H    Systemic...
CORTICOS... 14827
7 H02    1991 Aug 16519 General      Co-paym... H    Systemic...
CORTICOS... 3397
8 H02    1991 Aug 13894 General      Safety ... H    Systemic...
CORTICOS... 2970
9 H02    1991 Sep 269703 Concessional Co-paym... H    Systemic...
CORTICOS... 52822
10 H02    1991 Sep 128900 Concessional Safety ... H    Systemic...
CORTICOS... 18896
# i 806 more rows

```

```

PBS_H02_monthly <- PBS |>
  filter(ATC2 == "H02") |>
  index_by(Month) |>
  summarise(total_cost = sum(Cost, na.rm = TRUE))

```

```
PBS_H02_monthly
```

```

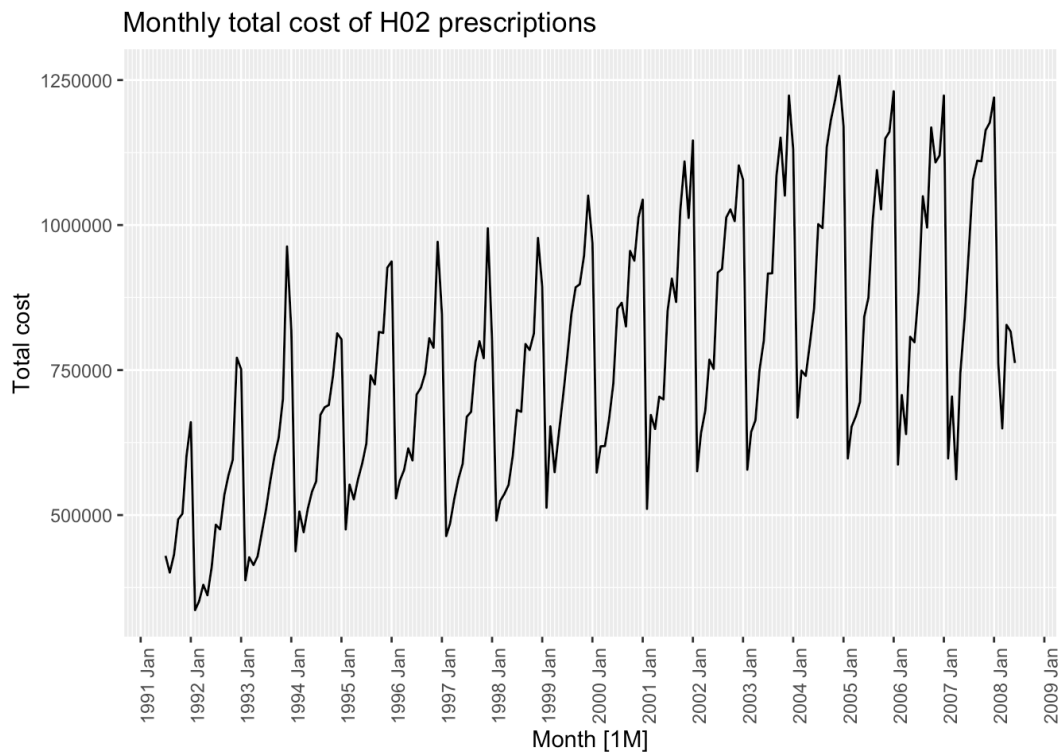
# A tsibble: 204 x 2 [1M]
      Month total_cost
      <mtl>      <dbl>
1 1991 Jul      429795
2 1991 Aug      400906
3 1991 Sep      432159
4 1991 Oct      492543
5 1991 Nov      502369
6 1991 Dec      602652
7 1992 Jan      660119
8 1992 Feb      336220
9 1992 Mar      351348
10 1992 Apr      379808
# i 194 more rows

```

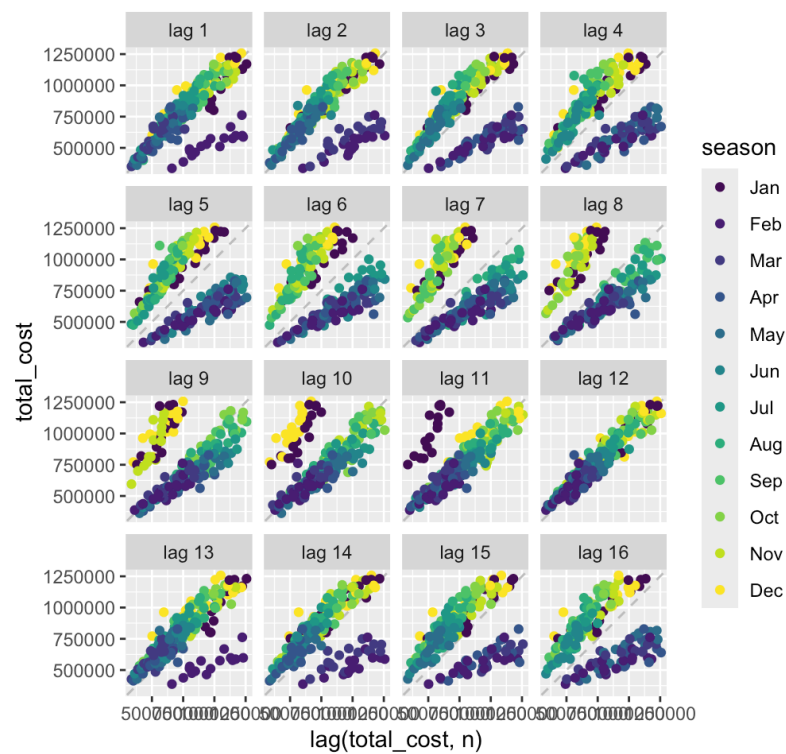
```

PBS_H02_monthly |>
  autoplot(total_cost) +
  scale_x_yearmonth(
    date_breaks = "1 year",
    date_minor_breaks = "1 month"
  ) +
  theme(axis.text.x = element_text(angle = 90)) +
  labs(
    title = "Monthly total cost of H02 prescriptions",
    y = "Total cost"
  )

```



```
PBS_H02_monthly |>
  gg_lag(total_cost, geom = "point", lags = 1:16)
```



```
PBS_H02_monthly |>
  ACF(total_cost) |>
```

ACF of monthly H02 prescription cost

acf

lag [1M]

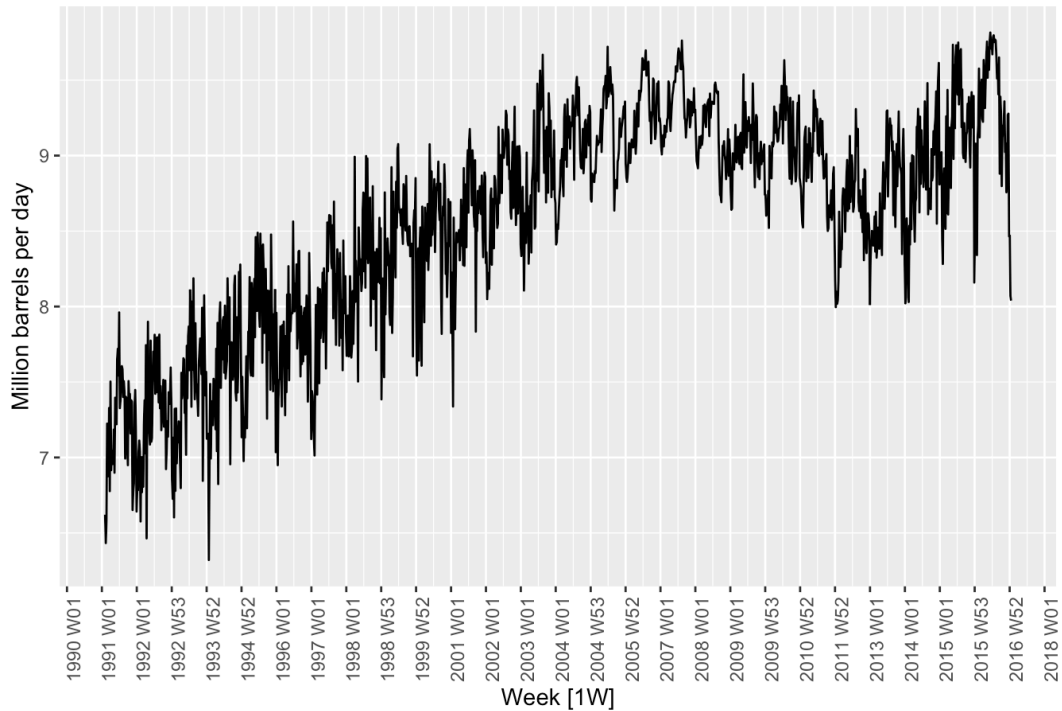
```
head(us_gasoline)
```

```
us_gasoline |>
  autoplot(Barrels) +
  scale_x_yearweek(
    date_breaks = "1 year"
  ) +
  theme(axis.text.x = element_text(angle = 90)) +
  labs(
    title = "Weekly US gasoline supply",

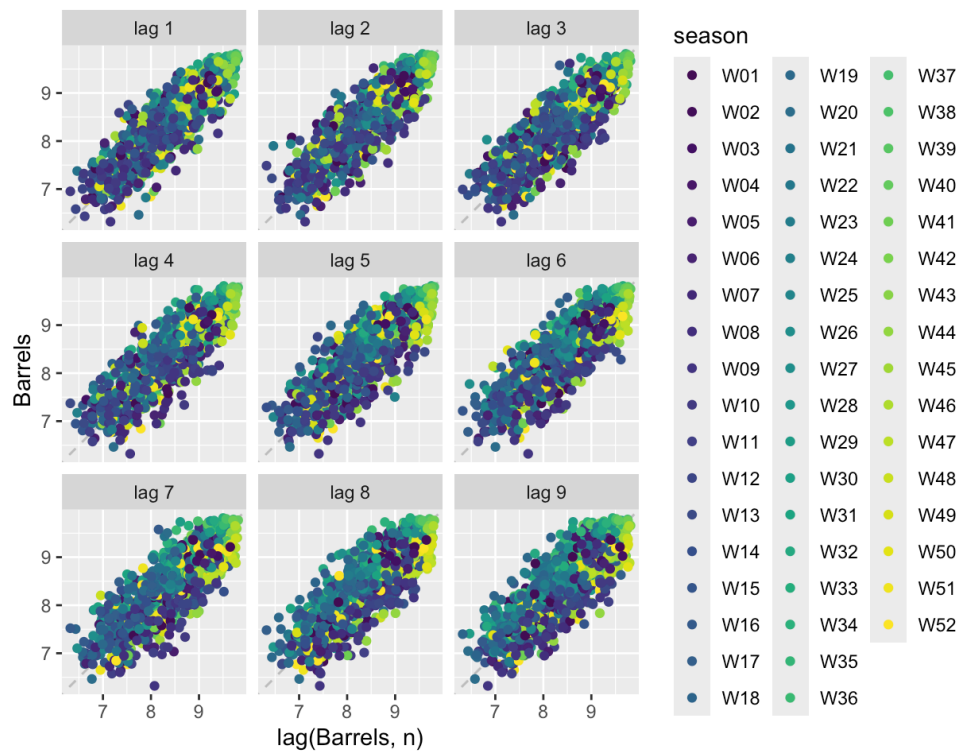
```

```
y = "Million barrels per day"
)
```

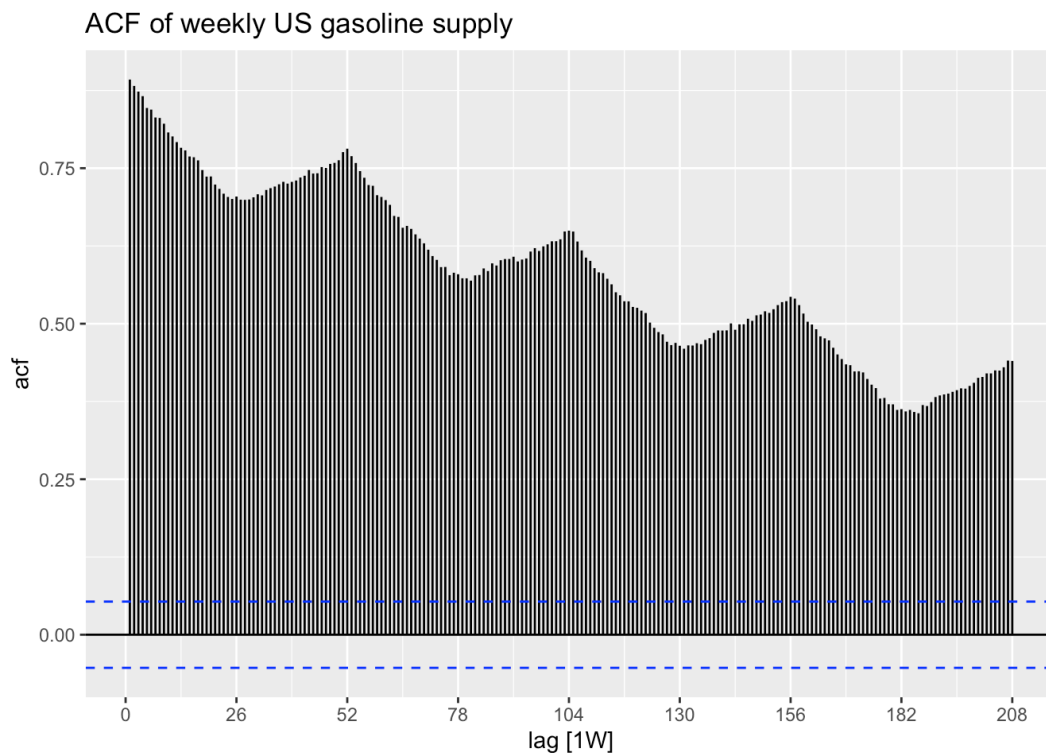
Weekly US gasoline supply



```
us_gasoline |>
  gg_lag(Barrels, geom = "point")
```



```
us_gasoline |>
  ACF(Barrels, lag_max = 52 * 4) |>
  autoplot() +
  labs(title = "ACF of weekly US gasoline supply")
```



```
#data is weekly, one yearly seasonal cycle is about 52 lags, so
```

Time plot: Gasoline supply has repeated up-and-down movement within each year, so it suggests possible yearly seasonality
 Lag plots: The lag plots show positive relationships at nearby lags: points slope upward, meaning weeks close together tend to have similar gasoline supply. But this plot is too crowded to clearly identify yearly seasonality by itself.
 ACF plot: Correlation rises very clearly every 52 weeks (a year), which makes seasonality evident

04_D_TSGraphs_MoreExercises

Exercise 1

```
pollution <-

# Read the file
read_csv("../data/Beijing_Pollution_TSeries.csv") |>

#Step 1: create the time_stamp
```

```
mutate(time_stamp = make_datetime(year, month, day, hour)) |>

#Step 2: reorder columns and drop timestamp components (keep c
select(time_stamp, everything(), -c(No, year, month, day, hour

#Step 3: convert to tsibble
as_tsibble(index = time_stamp)
```

Rows: 35064 Columns: 18

— Column specification

Delimiter: ","

chr (2): wd, station

dbl (16): No, year, month, day, hour, PM2.5, PM10, S02, N02, C0, 03, TEMP, P...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

pollution

A tsibble: 35,064 x 14 [1h] <UTC>

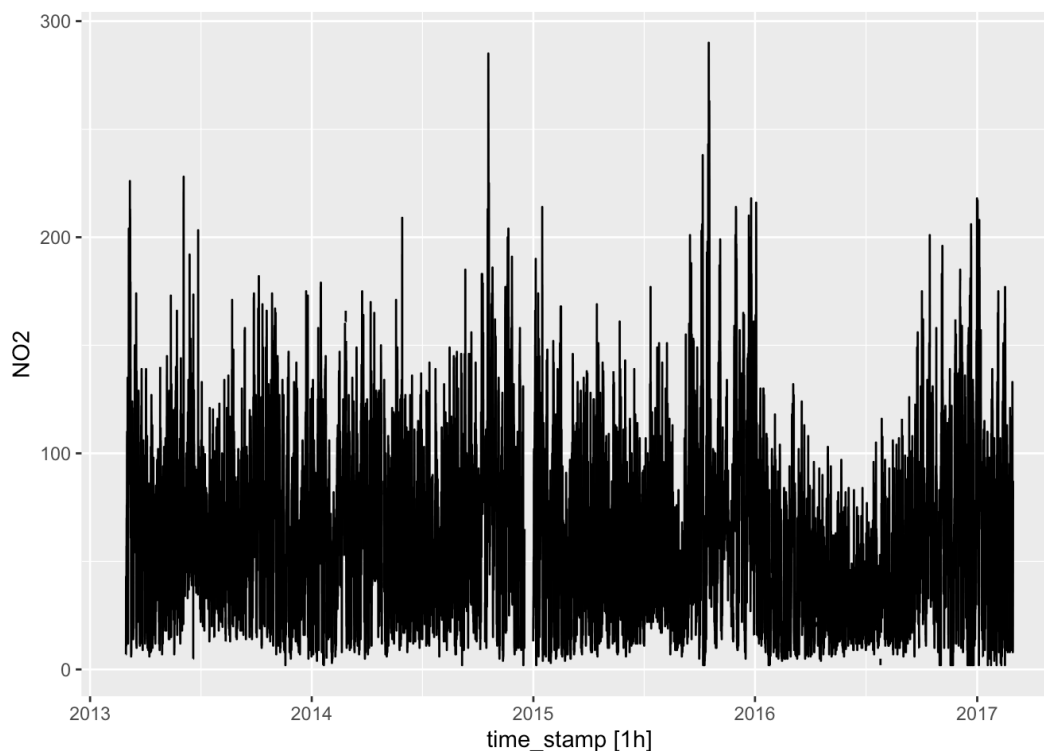
time_stamp	PM2.5	PM10	S02	N02	C0	03
TEMP PRES DEWP						
<dtm>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
<dbl> <dbl> <dbl>						
1 2013-03-01 00:00:00	4	4	4	7	300	77
-0.7 1023 -18.8						
2 2013-03-01 01:00:00	8	8	4	7	300	77
-1.1 1023. -18.2						
3 2013-03-01 02:00:00	7	7	5	10	300	73
-1.1 1024. -18.2						
4 2013-03-01 03:00:00	6	6	11	11	300	72
-1.4 1024. -19.4						
5 2013-03-01 04:00:00	3	3	12	12	300	72
1025. -19.5						-2
6 2013-03-01 05:00:00	5	5	18	18	400	66
-2.2 1026. -19.6						
7 2013-03-01 06:00:00	3	3	18	32	500	50
-2.6 1026. -19.1						
8 2013-03-01 07:00:00	3	6	19	41	500	43
-1.6 1027. -19.1						
9 2013-03-01 08:00:00	3	6	16	43	500	45
0.1 1028. -19.2						
10 2013-03-01 09:00:00	3	8	12	28	400	59

```
1.2 1028. -19.3
```

```
# i 35,054 more rows
```

```
# i 4 more variables: RAIN <dbl>, wd <chr>, WSPM <dbl>, station  
<chr>
```

```
autoplot(pollution, N02)
```



```
# it looks like there is seasonal variance but it's hard to tell
```

```
N02_weekly <- pollution |>  
  index_by(week = yearweek(time_stamp)) |>  
  summarise(N02_weekly = mean(N02, na.rm = TRUE)) |>  
  mutate(year_month = yearmonth(week)) |>  
  filter(year_month >= yearmonth("2013 Mar") &  
         year_month < yearmonth("2014 Mar"))
```

```
N02_weekly
```

```
# A tsibble: 52 x 3 [1W]
```

	week	N02_weekly	year_month
	<week>	<dbl>	<mth>
1	2013 W10	97.8	2013 Mar
2	2013 W11	82.0	2013 Mar
3	2013 W12	55.2	2013 Mar
4	2013 W13	72.2	2013 Mar

```

5 2013 W14      60.0    2013 Apr
6 2013 W15      44.7    2013 Apr
7 2013 W16      55.1    2013 Apr
8 2013 W17      59.0    2013 Apr
9 2013 W18      45.6    2013 Apr
10 2013 W19     65.4    2013 May
# i 42 more rows

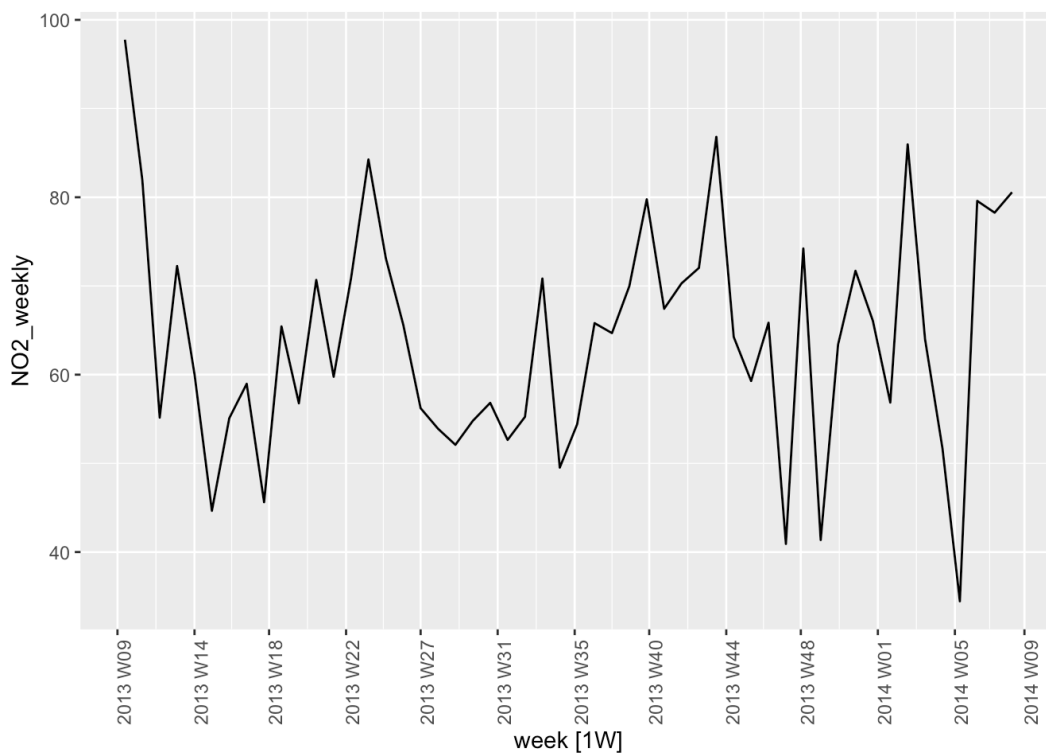
```

Exercise 1.2 - plotting

```

N02_weekly |>
  autoplot(N02_weekly) +
  scale_x_yearweek(
    date_breaks = "1 month"
  ) +
  theme(axis.text.x = element_text(angle = 90))

```



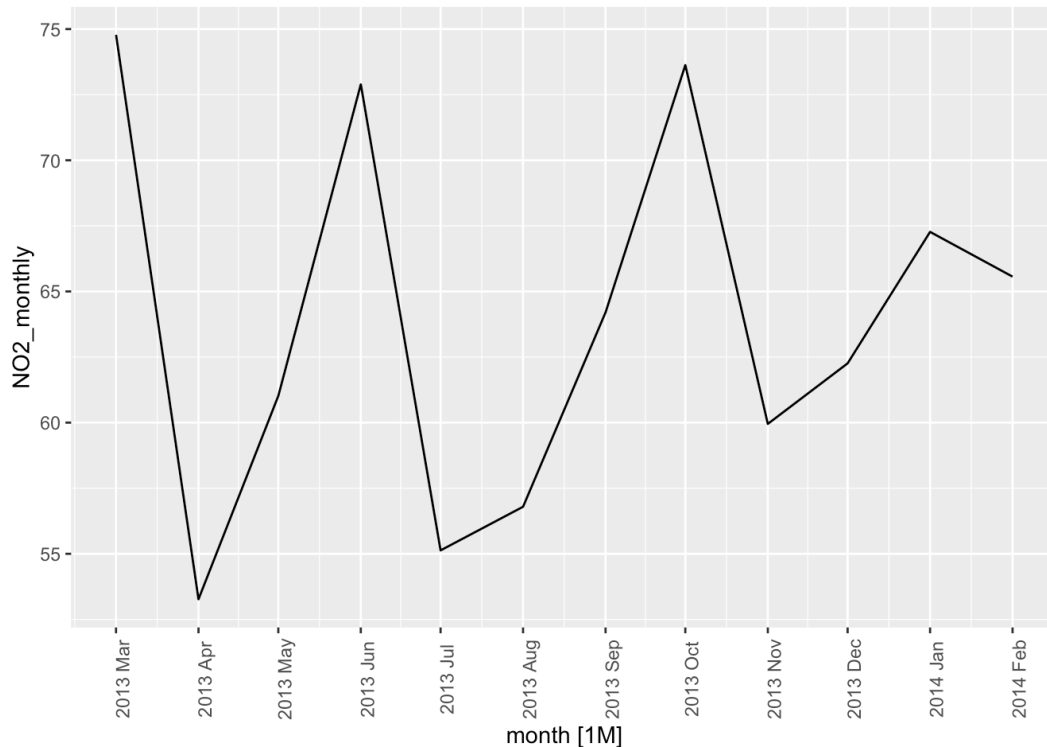
Weekly NO2 levels fluctuate strongly but there is no clear long-term trend only sharp spikes and drops

```

N02_monthly <- pollution |>
  index_by(month = yearmonth(time_stamp)) |>
  summarise(N02_monthly = mean(N02, na.rm = TRUE)) |>
  mutate(year_month = month) |>
  filter(year_month >= yearmonth("2013 Mar") &
    year_month < yearmonth("2014 Mar"))

```

```
N02_monthly |>
  autoplot(N02_monthly) +
  scale_x_yearmonth(
    date_breaks = "1 month"
  ) +
  theme(axis.text.x = element_text(angle = 90))
```



There is a slight overall upward trend in pollution, with around every 3-4 months having a spike.

Exercise 2

```
fuel_prices <-

# Read excel file
read_xlsx("../data/Weekly Fuel Prices.xlsx",
  col_types = c("date", "numeric", "numeric")) |>

filter(!is.na(Date)) |>

pivot_longer(cols = c("Petrol (USD)", "Diesel (USD)"),
  names_to = "fuel_type",
  values_to = "price_USD"
) |>

as_tsibble(key = fuel_type, regular = FALSE, index = Date)
```

Warning: Expecting date in A884 / R884C1: got 'Return to Contents page'

```
fuel_prices
```

```
# A tsibble: 1,762 x 3 [!] <UTC>
# Key:      fuel_type [2]
  Date                fuel_type    price_USD
  <dtm>              <chr>        <dbl>
1 2003-06-09 00:00:00 Diesel (USD)    76.8
2 2003-06-16 00:00:00 Diesel (USD)    76.7
3 2003-06-23 00:00:00 Diesel (USD)    76.6
4 2003-06-30 00:00:00 Diesel (USD)    76.5
5 2003-07-07 00:00:00 Diesel (USD)    76.5
6 2003-07-14 00:00:00 Diesel (USD)    76.4
7 2003-07-21 00:00:00 Diesel (USD)    76.9
8 2003-07-28 00:00:00 Diesel (USD)    76.9
9 2003-08-04 00:00:00 Diesel (USD)    76.8
10 2003-08-11 00:00:00 Diesel (USD)    77.1
# i 1,752 more rows
```

```
fuel_quarterly <-

  fuel_prices |>

  # Use group_by so that the subsequent index_by is applied
  # group-wise.
  group_by(fuel_type) |>

  # Index by yearquarter
  index_by(yq = yearquarter(Date)) |>

  # Summarise
  summarise(
    # One mean per fuel_type per quarter, so the name must not
    mean_price_USD = mean(price_USD, na.rm = TRUE)
  ) |>

  ungroup()

fuel_quarterly
```

```
# A tsibble: 138 x 3 [1Q]
# Key:      fuel_type [2]
  fuel_type    yq mean_price_USD
  <chr>        <qtr>        <dbl>
1 Diesel (USD) 2003 Q2          76.6
```

```

2 Diesel (USD) 2003 Q3          77.2
3 Diesel (USD) 2003 Q4          77.6
4 Diesel (USD) 2004 Q1          78.2
5 Diesel (USD) 2004 Q2          81.5
6 Diesel (USD) 2004 Q3          82.2
7 Diesel (USD) 2004 Q4          85.7
8 Diesel (USD) 2005 Q1          85.1
9 Diesel (USD) 2005 Q2          89.2
10 Diesel (USD) 2005 Q3         94.5
# i 128 more rows

```

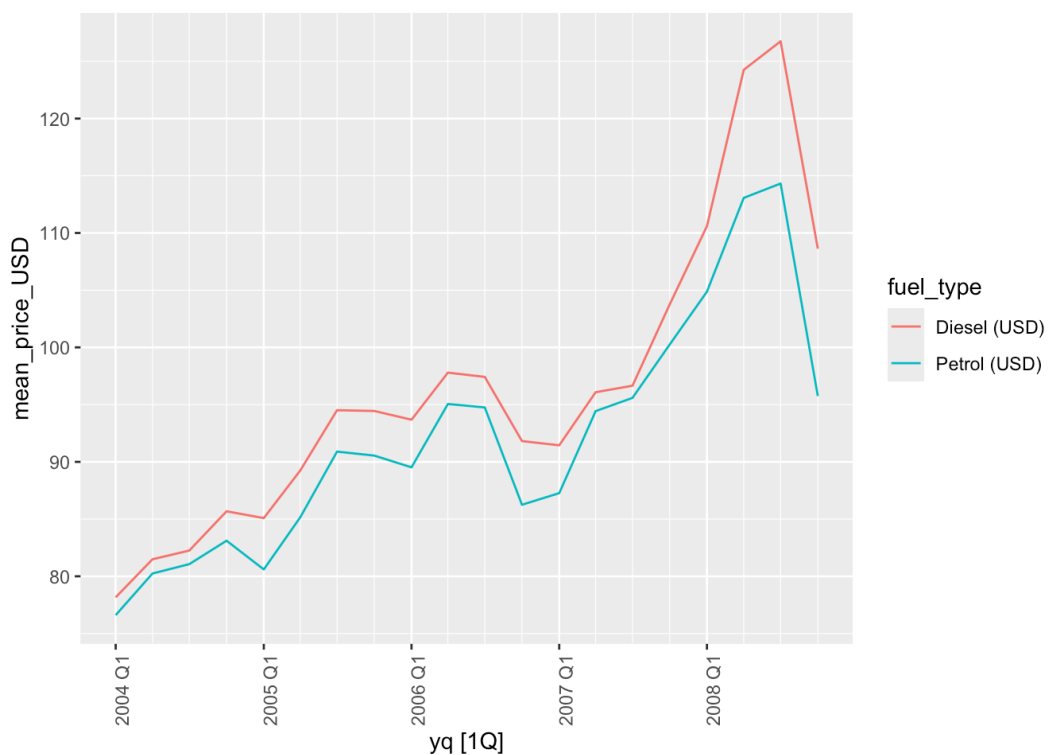
With `group_by(fuel_type)` there is quarterly data both for petrol and diesel separately. Without it they are averaged together into one combined series.

```

fuel_quarterly_2004_2008 <- fuel_quarterly |>
  filter(year(yq) >= 2004 & year(yq) <= 2008)

fuel_quarterly_2004_2008 |>
  autoplot(mean_price_USD) +
  scale_x_yearquarter(
    date_breaks = "1 year",
    minor_breaks = "3 months"
  ) +
  theme(axis.text.x = element_text(angle = 90))

```



First of all we can see that diesel and petrol prices are highly correlated

and move almost perfectly together. Secondly, there has been a strong trend of price increase form 2004 until 2008 Q3, which was followed by a sharp drop (likely due to 2008 financial crisis).